# Homework Set B.1

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## Problem 1

Let $v$ be the vector to the corner at $(1,1)$, so $v=(1,1)$. The $x$-axis is a line spanned by $\hat{i}=(1,0)$. Let $\theta \in \mathbb{R}$ be the angle between vectors $x$ and $\hat{i}$. By definition of the dot product:

$$v \cdot \hat{i} = 1 = \|v\|\|\hat{i}\| \cos\theta = \sqrt{2}(1)\cos\theta \implies \cos\theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \implies \theta = \frac{\pi}{4} = \boxed{45^\circ}$$

## Problem 2

Let $v$ be the vector from the origin to the corner, so $v=(1,1,1)$. The $x$-axis is spanned by $\hat{i}=(1,0,0)$.

$$v \cdot \hat{i} = 1 = \|v\|\|\hat{i}\| \cos\theta = \sqrt{3}(1)\cos\theta \implies \cos\theta = \frac{1}{\sqrt{3}} \implies \theta = \arccos\left(\frac{1}{\sqrt{3}}\right) \approx \boxed{54.7^\circ}$$

## Problem 3

$v=(1,1,1,1), \hat{i}=(1,0,0,0)$.

$$v \cdot \hat{i} = 1 = \|v\|\|\hat{i}\| \cos\theta = \sqrt{4}(1)\cos\theta \implies \cos\theta = \frac{1}{2} \implies \theta = \frac{\pi}{3} = \boxed{60^\circ}$$

## Problem 4

In $\mathbb{R}^n$, $v \cdot \hat{i} = 1$, $\|\hat{i}\| = 1$, and the magnitude of $v$ is:

$$\|v\| = \sqrt{\sum_{i=1}^{n} v_i^2} = \sqrt{\sum_{i=1}^{n} 1} = \sqrt{n}$$

Let $\theta$ be the angle between $v$ and $\hat{i}$. By definition of the dot product:

$$v \cdot \hat{i} = 1 = \|v\|\|\hat{i}\| \cos\theta = \sqrt{n}\cos\theta \implies \cos\theta = \frac{1}{\sqrt{n}}$$

As $n \to \infty$, $\sqrt{n} \to \infty$ and $1/\sqrt{n} \to 0$.

$$\cos\theta = 0 \implies \theta = \frac{\pi}{2} = \boxed{90^\circ}$$

The angle between the $x$-axis and the vector to one vertex of the hypercube tends to $90^\circ$ as the number of dimensions $n \to \infty$.

## Problem 5

Let $\hat{u}$ and $\hat{v}$ be unit vectors in the same direction as $u$ and $v$, respectively.

$$\hat{u} = \frac{u}{\|u\|} \qquad \hat{v} = \frac{v}{\|v\|} \qquad w' = \hat{u} + \hat{v}$$

Since $\hat{u}$ and $u$ point in the same direction, $w'$ bisects $u$ and $v$ if and only if it bisects $\hat{u}$ and $\hat{v}$. Let $\theta \in [0,\pi]$ be angle between $\hat{u}$ and $w'$, and let $\varphi \in [0,\pi]$ be the angle between $\hat{v}$ and $w'$.

$$\hat{u} \cdot w' = \|\hat{u}\|\|w'\| \cos\theta \iff \hat{u} \cdot (\hat{u} + \hat{v}) = \|w'\| \cos\theta \iff \cos\theta = \frac{\hat{u}\cdot\hat{u} + \hat{u}\cdot\hat{v}}{\|w'\|} = \frac{1 + \hat{u}\cdot\hat{v}}{\|w'\|}$$

$$\hat{v} \cdot w' = \|\hat{v}\|\|w'\| \cos\varphi \iff \hat{v} \cdot (\hat{u} + \hat{v}) = \|w'\| \cos\varphi \iff \cos\varphi = \frac{\hat{v}\cdot\hat{u} + \hat{v}\cdot\hat{v}}{\|w'\|} = \frac{1 + \hat{u}\cdot\hat{v}}{\|w'\|}$$

Cosine is injective on $[0,\pi]$ so $\cos\theta = \cos\varphi \implies \theta = \varphi$. Since the angle between $u$ and $w'$ is the same as the angle between $v$ and $w'$, the vector $w' = \hat{u} + \hat{v}$ bisects the angle between $u$ and $v$.

Notice that:

$$\|u\|\|v\|w' = \|u\|\|v\|\left(\frac{u}{\|u\|} + \frac{v}{\|v\|}\right) = \|v\|u + \|u\|v = w$$

Since $\|u\|\|v\|$ is a scalar, $w$ and $w'$ point in the same direction. $w$ bisects the angle between $u$ and $v$. $\square$